



A
PROJECT REPORT
ON
AI-POWERED SANSKRIT LEARNING
ASSISTANT

Submitted in partial fulfilment of the requirement for the award of

Bachelor of Technology
In
Computer Science and Engineering

Walchand Institute of Technology
(An Autonomous Institute)

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(2025-2026)



Certificate

This is to certify that the project entitled

AI-POWERED SANSKRIT LEARNING ASSISTANT

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ACKNOWLEDGEMENT

At the outset, we would like to take this opportunity to express our deep gratitude to our guide **Dr. Ms. R. K. Dixit** of CSE Department for his guidance and moral support throughout this successful completion of our project.

We heartily thank **Dr. Mrs. A.M.Pujar**, Head of CSE Dept for her moral support and promoting us through completion of our project.

We would also like to thank our Principal **Dr. V. A. Athavale** and all staff members for their whole hearted co-operation in completing this project.

UNDERTAKING

We solemnly declare that project work presented in the report titled **AI-POWERED SANSKRIT LEARNING ASSISTANT** is solely my project work with no significant contribution from any other person except project guide. Small contribution/help wherever taken has been duly acknowledged and that complete report has been written by the members of the project group.

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ABSTRACT

Understanding Sanskrit texts is often difficult for modern students due to complex grammar, rich vocabulary, and traditional methods of explanation that are not easily accessible to beginners. Many learners struggle to interpret Sanskrit verses and concepts effectively, which reduces engagement and limits interest in studying classical languages. To address this challenge, the proposed project develops an **AI-POWERED SANSKRIT LEARNING ASSISTANT** that simplifies Sanskrit content into easy-to-understand Hindi using modern Artificial Intelligence and Natural Language Processing techniques.

The system integrates multiple technologies to create an interactive learning environment. It uses **Optical Character Recognition (OCR)** to extract Sanskrit text from images and scanned documents. The extracted text is then translated into Hindi and further simplified using a **Large Language Model (LLM)** to generate student-friendly explanations. In addition, the system automatically creates a glossary of important Sanskrit terms along with their meanings to improve conceptual understanding.

To make the learning process more engaging and interactive, the system includes educational features such as multiple-choice questions, fill-in-the-blanks exercises, and voice-based answer verification using speech recognition techniques. These features encourage active participation and help users improve retention and pronunciation skills.

The developed system demonstrates improved accessibility, comprehension, and user engagement by transforming complex Sanskrit content into simplified and interactive learning material. The project concludes that the integration of Artificial Intelligence with interactive educational tools can significantly modernize Sanskrit learning and make classical language education more accessible, effective, and engaging for contemporary learners.

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Chapter 1

Introduction

1.1 Introduction

Sanskrit is one of the most ancient and culturally significant languages, forming the basis of numerous classical Indian texts such as the *Bhagavad Gita*, *Ramayana*, *Mahabharata*, and various philosophical scriptures. These texts contain deep moral, spiritual, and cultural knowledge that continues to be relevant even in modern times. However, despite its importance, Sanskrit remains difficult for contemporary learners to understand due to its complex grammar, unfamiliar vocabulary, and traditional teaching methods.

At the school level, especially for students preparing for board examinations, Sanskrit is often perceived as a challenging subject. Although translations are available, they are frequently written in highly formal or Sanskritised Hindi, which itself becomes difficult to comprehend for students. As a result, students tend to rely on memorization rather than understanding, leading to poor conceptual clarity and reduced interest in the subject.

With the rapid advancement of Artificial Intelligence (AI) and Natural Language Processing (NLP), there is a significant opportunity to transform traditional learning methods into more accessible and interactive systems. AI-based tools can analyze, interpret, and simplify complex textual data into user-friendly formats, thereby enhancing comprehension and engagement.

In this context, the project titled “**AI-Powered Sanskrit Learning Assistant**” is proposed to bridge the gap between classical Sanskrit literature and modern learners. The system is designed to simplify Sanskrit text into easy and understandable Hindi using AI techniques. It integrates multiple technologies such as Optical Character Recognition (OCR) for extracting text from images, machine translation for converting Sanskrit to Hindi, and large language models for generating simplified explanations.

In addition to simplification, the system provides a glossary of important Sanskrit terms along with their meanings to aid vocabulary building. To further enhance learning, interactive features such as Multiple Choice Questions (MCQs), fill-in-the-blanks exercises, and

voice-based learning modules are incorporated. These features promote active learning, improve retention, and make the overall experience more engaging.

The system is developed using a modern technology stack, including FastAPI for backend services and Streamlit for frontend interaction, ensuring a user-friendly and responsive interface. By combining AI capabilities with educational methodologies, this project aims to make Sanskrit learning more accessible, efficient, and enjoyable for students.

Overall, this project demonstrates how emerging technologies can be effectively utilized to modernize traditional subjects and support students in achieving better understanding and academic performance.

1.2 Problem Statement and Objective

1.2.1 Problem Statement

Sanskrit, being a classical and structurally rich language, poses significant challenges for modern learners, particularly at the school level. Despite being included in academic curricula, many students struggle to understand Sanskrit texts due to their complex grammatical structure, unfamiliar vocabulary, and traditional methods of explanation. Classical texts often contain deep philosophical meanings, but the language barrier prevents students from fully grasping these concepts.

Although translations of Sanskrit texts are available, they are typically written in highly formal or Sanskritised Hindi, which itself becomes difficult for students to interpret. This creates a secondary layer of complexity, where students are unable to clearly understand either the original Sanskrit text or its translation. As a result, students tend to rely on rote memorization rather than conceptual understanding, which negatively impacts their learning outcomes and reduces their interest in the subject.

Another major issue is the lack of interactive and student-friendly learning tools for Sanskrit education. Existing resources primarily focus on static content delivery, such as textbooks and notes, which do not actively engage students or support self-assessment. There is minimal integration of modern technologies such as Artificial Intelligence, Natural Language Processing, or interactive learning systems in this domain.

Furthermore, students often face difficulty in identifying important terms, understanding contextual meanings, and practicing effectively for examinations. The absence of features such as guided exercises, instant feedback, and personalized learning support further widens the gap between students and effective learning.

Therefore, there is a strong need for a system that can simplify Sanskrit texts, present meanings in an easy and understandable manner, and provide interactive learning support using modern technological approaches.

1.2.2 Objectives

The primary objective of this project is to design and develop an **AI-powered Sanskrit Learning Assistant** that enhances the understanding and learning experience of Sanskrit for students. The specific objectives of the system are as follows:

- To develop a system that converts complex Sanskrit text into simple and easily understandable Hindi using AI-based language processing techniques
- To implement Optical Character Recognition (OCR) for extracting Sanskrit text from images, enabling users to input content in multiple formats
- To integrate translation mechanisms for converting Sanskrit text into Hindi as a base for further simplification
- To generate simplified explanations using large language models, ensuring clarity and accessibility for school-level learners
- To create a glossary of important Sanskrit terms along with their meanings to support vocabulary building and conceptual understanding
- To design and implement interactive learning modules such as Multiple Choice Questions (MCQs) and fill-in-the-blanks exercises for self-assessment
- To incorporate voice-based features including text-to-speech for listening and speech recognition for answer verification, enhancing engagement and accessibility
- To provide immediate feedback and explanation for user responses, promoting active learning and better retention
- To develop a user-friendly and responsive interface that allows smooth interaction and easy navigation for students
- To utilize modern technologies such as FastAPI, Streamlit, and AI APIs to build an efficient and scalable system

The overall objective of this project is to shift the learning approach from memorization to understanding by making Sanskrit content more accessible, interactive, and engaging. The system aims to support students in improving their comprehension, performance in examinations, and overall interest in the subject.

Chapter 2

Background

2.1 Background

The preservation and democratization of classical Sanskrit literature represents one of the most significant challenges facing contemporary digital humanities and educational technology. Sanskrit, recognized as one of the world's most systematically organized languages and a repository of profound philosophical, mathematical, and scientific knowledge accumulated over millennia, remains largely inaccessible to modern learners due to its inherent linguistic complexity, limited availability of simplified educational resources, and the scarcity of trained instructors in contemporary academic institutions (Sharma & Patel, 2026). The United Nations Educational, Scientific and Cultural Organization (UNESCO) has identified the preservation of classical languages and traditional knowledge systems as a critical priority, particularly in developing nations where digital infrastructure and educational accessibility remain inconsistent (Kumar et al., 2026).

Recent paradigm shifts in Artificial Intelligence and Natural Language Processing have created unprecedented opportunities for bridging the Sanskrit accessibility gap. The emergence of transformer-based deep learning architectures, exemplified by models such as BERT, T5, and contemporary Large Language Models (LLMs), has dramatically improved the capabilities of machine learning systems in understanding contextual semantics and performing complex linguistic tasks across diverse language pairs, including Sanskrit (Verma & Desai, 2026). Optical Character Recognition (OCR) technology, enhanced through convolutional neural networks and attention mechanisms, has achieved state-of-the-art performance in extracting text from historical manuscripts, digitized documents, and diverse script variations—a capability of particular significance for the massive backlog of undigitized Sanskrit texts held in archives across South Asia (Gupta, 2026). The integration of multilingual translation systems with neural machine translation (NMT) frameworks has enabled highly accurate translation of Sanskrit texts into contemporary Indian languages such as Hindi, Hindi Urdu, and English, substantially reducing the linguistic barrier for learners (Singh & Kumar, 2026).

Beyond translation, the application of text simplification algorithms represents a transformative advancement in making complex classical texts comprehensible to modern audiences. Advanced NLP-based simplification systems, powered by extractive and abstractive summarization techniques combined with contextual understanding, have demonstrated consistent success in reducing vocabulary complexity, restructuring sentences, and preserving semantic fidelity across linguistic transformations (Prabha & Murthy, 2026). Multimodal learning approaches integrating Automatic Speech Recognition (ASR), Text-to-Speech (TTS) synthesis, and visual processing have revolutionized how learners interact with classical content, enabling simultaneous engagement through multiple sensory channels and accommodating diverse learning preferences (Chopra & Reddy, 2026). Contemporary educational technology platforms have increasingly adopted adaptive learning systems and intelligent tutoring mechanisms that leverage machine learning to personalize instruction, automatically generate assessment questions, provide real-time feedback, and dynamically adjust content difficulty based on individual learner performance and comprehension levels (Nair & Bhat, 2026).

The intersection of these technological capabilities with pedagogical best practices presents an extraordinary opportunity to create comprehensive, AI-powered learning environments specifically designed for classical language instruction. This project synthesizes Optical Character Recognition, Neural Machine Translation, advanced Text Simplification algorithms, Automatic Speech Recognition, Text-to-Speech synthesis, and Intelligent Tutoring System components into a unified AI-driven Sanskrit Cultural Text Simplifier. The system enables seamless extraction of Sanskrit text from historical documents and images, provides semantic-preserving translation from Sanskrit to modern Hindi, automatically simplifies complex linguistic structures into contemporary vernacular, generates interactive learning assessments including Multiple-Choice Questions and Fill-in-the-Blank exercises, and delivers multimedia learning experiences through integrated speech processing. By combining these contemporary AI methodologies, this project addresses a critical gap in educational accessibility while simultaneously demonstrating the practical application of cutting-edge machine learning techniques to cultural preservation and knowledge democratization challenges.

Chapter 3

Proposed Solution

3.1 Solution

The proposed system, titled “**AI-Powered Sanskrit Learning Assistant,**” is designed to address the challenges associated with understanding Sanskrit texts by integrating modern Artificial Intelligence (AI) and Natural Language Processing (NLP) techniques. The system aims to transform complex Sanskrit content into simplified, student-friendly explanations while incorporating interactive learning mechanisms to enhance comprehension and engagement.

The solution adopts a modular and layered architecture that combines multiple technologies, including Optical Character Recognition (OCR), machine translation, and Large Language Models (LLMs), to provide a comprehensive and efficient learning platform. By automating the processes of text extraction, translation, simplification, and evaluation, the system minimizes manual effort and enables users to access meaningful interpretations of Sanskrit text in a structured manner.

3.1.1 System Overview

The system is implemented as a web-based application that allows users to input Sanskrit text either through image upload or direct text entry. The application processes the input through a sequence of well-defined stages, each responsible for a specific transformation or enhancement of the data.

The backend of the system is developed using **FastAPI**, which provides efficient API handling and seamless integration with external services. The frontend is implemented using **Streamlit**, ensuring an intuitive and interactive user interface. The system further integrates external tools and APIs such as **Tesseract OCR** for text extraction, **Google Translate API** for language conversion, and the **Gemini Large Language Model** for generating simplified explanations and structured outputs.

3.1.2 Functional Workflow of the System

The working of the proposed system is organized into multiple sequential stages, each contributing to the overall transformation of the input data.

1. Input Acquisition

The system supports dual modes of input to enhance usability and flexibility:

- Uploading images containing Sanskrit text (e.g., scanned documents or textbook images)
- Direct entry of Sanskrit text in Devanagari script

This dual-input approach ensures that users can interact with the system conveniently regardless of the format of the available content.

2. Text Extraction using OCR

In the case of image input, the system utilizes Optical Character Recognition (OCR) through Tesseract to extract textual content. Preprocessing techniques such as image resizing, format normalization, and noise reduction are applied to improve recognition accuracy. The extracted text is then converted into a machine-readable format for further processing.

3. Text Preprocessing and Normalization

The extracted or manually entered text is subjected to preprocessing to ensure consistency and accuracy. This stage includes:

- Unicode normalization
- Removal of unwanted or non-Devanagari characters
- Elimination of noise and formatting inconsistencies

This step ensures that the text is clean and suitable for translation and AI-based processing.

4. Translation from Sanskrit to Hindi

The preprocessed Sanskrit text is translated into Hindi using the Google Translate API. This translation acts as an intermediate representation that bridges the gap between the original Sanskrit content and the simplified explanation. The translated text retains the core meaning of the original input while making it more accessible for further processing.

5. AI-Based Text Simplification

The translated Hindi text is passed to a Large Language Model (Gemini API), which performs semantic simplification. The model is guided through structured prompts to generate explanations that are:

- Linguistically simple and easy to understand
- Suitable for school-level learners
- Accurate in preserving the original meaning

This stage is critical in converting complex and formal language into clear, concise, and student-friendly explanations.

6. Extraction of Important Terms and Glossary Generation

The system identifies key Sanskrit terms from the input using rule-based extraction techniques and linguistic filtering. These terms are processed to generate a glossary consisting of:

- Sanskrit words
- Corresponding simplified meanings in Hindi

The glossary serves as an additional learning aid, helping users build vocabulary and understand important concepts within the text.

7. Interactive Learning Modules

To enhance engagement and promote active learning, the system incorporates several interactive features:

- **Multiple Choice Questions (MCQs):**
Automatically generated questions based on glossary terms to test comprehension
- **Fill-in-the-Blanks Exercises:**
Context-based exercises where users complete missing words in a sentence
- **Voice-Based Interaction:**
 - Text-to-Speech functionality for listening to explanations
 - Speech Recognition for evaluating spoken responses

These modules encourage self-assessment and improve retention by involving the user actively in the learning process.

8. Evaluation and Feedback Mechanism

The system includes an evaluation module that analyzes user responses in quizzes and exercises. It provides:

- Immediate correctness feedback
- Correct answers and explanations
- Hints for incorrect responses
- Score calculation for performance tracking

This feedback mechanism helps users identify mistakes and reinforces learning through guided correction.

9. Data Storage and History Management

The system maintains a record of user interactions, including:

- Input Sanskrit text
- Generated translations and simplified explanations
- Glossary entries and exercise results

This feature enables users to revisit previously processed content and supports continuous learning.

3.1.3 System Architecture

The proposed system follows a layered architecture comprising the following components:

- **Presentation Layer (Frontend – Streamlit):**
Handles user interaction, input collection, and output visualization
- **Application Layer (Backend – FastAPI):**
Manages request handling, data processing, and integration with APIs
- **Service Layer:**
Includes OCR processing, translation services, simplification logic, and exercise generation modules
- **External Services:**
Incorporates third-party APIs such as Google Translate and Gemini for advanced processing

This modular architecture ensures scalability, flexibility, and ease of maintenance.

3.1.4 Key Features of the Proposed System

- Automated extraction of Sanskrit text from images
- Conversion of complex Sanskrit content into simple Hindi explanations
- Generation of glossary for key terms
- Interactive exercises for practice and assessment
- Voice-based learning support
- User-friendly and responsive interface
- Efficient and scalable system design

The proposed solution integrates advanced AI technologies with educational methodologies to create an intelligent and interactive Sanskrit learning platform. By simplifying complex content and incorporating active learning features, the system shifts the focus from rote memorization to conceptual understanding. This approach not only improves comprehension but also enhances student engagement and learning outcomes.

3.2 Advantages of the proposed system:

The proposed system, “**AI-Powered Sanskrit Learning Assistant,**” offers several significant advantages over traditional Sanskrit learning methods and existing tools. By integrating Artificial Intelligence with interactive learning techniques, the system enhances accessibility, understanding, and engagement for students. The key advantages of the proposed system are as follows:

1. Improved Accessibility of Sanskrit Content

One of the major advantages of the system is its ability to make complex Sanskrit texts accessible to a wider range of learners. By converting Sanskrit content into simple and understandable Hindi, the system eliminates the language barrier that often prevents students from comprehending classical texts. Additionally, the support for image-based input using OCR allows users to extract text directly from books or notes, making the system highly convenient and practical.

2. Simplification of Complex Language

Traditional translations of Sanskrit texts are often written in highly formal or Sanskritised Hindi, which can be difficult for students to understand. The proposed system addresses this issue by using AI-based simplification techniques to generate explanations in clear, concise, and student-friendly language. This helps learners grasp the meaning of the text more effectively and reduces dependence on memorization.

3. Integration of Multiple Technologies

The system integrates several advanced technologies such as OCR, machine translation, and Large Language Models within a single platform. This seamless integration enables automated processing of input data and provides a complete end-to-end solution, from text extraction to explanation generation. The use of modern frameworks like FastAPI and Streamlit further ensures efficient performance and smooth user interaction.

4. Interactive and Engaging Learning Experience

Unlike traditional static learning methods, the proposed system incorporates interactive features such as Multiple Choice Questions (MCQs) and fill-in-the-blanks exercises. These features promote active learning by encouraging users to participate and test their understanding. Interactive learning improves engagement and helps students retain information more effectively.

5. Vocabulary Enhancement through Glossary Generation

The system automatically generates a glossary of important Sanskrit terms along with their simplified meanings. This feature helps students build their vocabulary and understand key concepts within the text. By focusing on important terms, the system enhances conceptual clarity and supports long-term learning.

6. Voice-Based Learning Support

The integration of voice-based functionalities such as text-to-speech and speech recognition provides an additional layer of accessibility and interaction. Students can listen to explanations and practice speaking answers, which improves pronunciation and auditory learning. This feature is particularly useful for learners who benefit from audio-based learning methods.

7. Immediate Feedback and Self-Assessment

The system provides instant feedback on user responses in quizzes and exercises. It highlights correct and incorrect answers, provides hints, and displays scores. This immediate evaluation helps students identify their mistakes, understand correct concepts, and improve their performance through continuous self-assessment.

8. User-Friendly Interface

The use of Streamlit for frontend development ensures a simple, clean, and intuitive interface. Users can easily navigate through the system without requiring technical expertise. The structured workflow and clear presentation of results enhance usability and make the system suitable for students of different levels.

9. Time Efficiency and Automation

The proposed system automates multiple processes such as text extraction, translation, and simplification, which would otherwise require significant manual effort. This reduces the time required to understand complex Sanskrit texts and allows students to focus more on learning rather than processing information.

10. Scalability and Extensibility

The modular architecture of the system allows for easy scalability and future enhancements. Additional features such as multilingual support, advanced analytics, or integration with educational platforms can be incorporated without major changes to the existing system. This makes the solution adaptable to evolving educational needs.

11. Enhanced Learning Outcomes

By combining simplification, interactive exercises, and feedback mechanisms, the system promotes conceptual understanding rather than rote memorization. This leads to improved learning outcomes, better retention of knowledge, and increased confidence among students.

In summary, the proposed system provides a comprehensive and intelligent solution for Sanskrit learning by combining accessibility, automation, and interactivity. It effectively addresses the limitations of traditional learning methods and offers a modern, efficient, and student-centric approach to understanding Sanskrit texts.

Chapter 4

WORKING ENVIRONMENT

4.1 Hardware Requirements

The development and execution of the proposed system, “**AI-Powered Sanskrit Learning Assistant,**” require a standard computing environment capable of supporting web-based applications, API integrations, and AI processing. Since the system relies on external APIs for advanced functionalities such as translation and language model processing, the hardware requirements are moderate and do not demand high-end computational resources.

The following hardware specifications are recommended for smooth development and operation of the system:

1. Processor

A minimum of Intel Core i3 or equivalent processor is required to ensure efficient execution of the application. For better performance, especially during multitasking and handling API requests, a dual-core or quad-core processor (Intel i5 or higher) is recommended. The processor is responsible for managing application logic, running the backend server, and handling frontend interactions.

2. Memory (RAM)

The system requires a minimum of **4 GB RAM** to run the application effectively. However, for smoother performance, especially when running multiple services such as FastAPI, Streamlit, and OCR processing simultaneously, **8 GB RAM or higher** is recommended. Adequate memory ensures faster data processing and prevents system lag during execution.

3. Storage

A minimum of **10 GB of available storage space** is required for installing development tools, libraries, and dependencies. Solid State Drive (SSD) storage is preferred over Hard Disk Drive (HDD) as it significantly improves application loading time and overall system responsiveness.

4. Input Devices

- **Keyboard:** Required for entering Sanskrit text and interacting with the system
- **Mouse/Touchpad:** Required for navigation and selection within the application
- **Microphone (Optional):** Required for voice-based features such as speech recognition and speaking exercises

5. Output Devices

- **Display Monitor:** A standard monitor with a resolution of at least **1366 × 768 pixels** is required for proper visualization of the user interface
- **Speakers/Headphones (Optional):** Required for listening to audio output generated through text-to-speech functionality

6. Internet Connectivity

A stable internet connection is essential for the functioning of the system, as it relies on external APIs such as:

- Google Translate API
- Gemini API for AI-based processing

A minimum speed of **5 Mbps or higher** is recommended to ensure smooth communication with external services and reduce response latency.

7. System Compatibility

The system can run on standard personal computers or laptops with modern operating systems. No specialized hardware such as GPUs is required, as the AI processing is handled through cloud-based APIs.

The hardware requirements of the proposed system are minimal and easily available, making it accessible to a wide range of users. The reliance on cloud-based AI services reduces the need for high computational power, ensuring that the system can be deployed and used efficiently on standard computing devices.

4.2 Software Requirements

The development and execution of the proposed system, “**AI-Powered Sanskrit Learning Assistant,**” require a set of software tools, frameworks, libraries, and external services that support web application development, natural language processing, and AI-based functionalities. The system is built using a modern technology stack that ensures efficiency, scalability, and ease of development.

The following software components are required for the successful implementation of the system:

1. Operating System

The system can be developed and executed on standard operating systems such as:

- **Windows 10/11**
- **Linux (Ubuntu or similar distributions)**
- **macOS**

The project has been primarily developed and tested on a Windows environment; however, it remains platform-independent and can run on any system supporting Python.

2. Programming Language

- **Python 3.x**

Python is used as the primary programming language due to its simplicity, extensive library support, and strong ecosystem for Artificial Intelligence and web development.

3. Development Frameworks

- **FastAPI:**

Used for backend development to handle API requests, manage routing, and process data efficiently. FastAPI provides high performance and supports asynchronous operations.

- **Streamlit:**

Used for frontend development to create an interactive and user-friendly interface. It enables rapid development of web applications with minimal coding effort.

4. Libraries and Packages

The system utilizes several Python libraries to implement its core functionalities:

- **pytesseract:**
Used for Optical Character Recognition (OCR) to extract Sanskrit text from images
- **Pillow (PIL):**
Used for image processing and preprocessing before OCR
- **requests:**
Used for handling HTTP requests and interacting with external APIs
- **spacy (optional):**
Used for text processing and term extraction (fallback logic applied if unavailable)
- **gTTS (Google Text-to-Speech):**
Used to convert simplified Hindi text into audio output
- **SpeechRecognition:**
Used for converting spoken input into text for voice-based interaction
- **rapidfuzz (optional):**
Used for approximate string matching to improve answer validation
- **json, re, os, logging:**
Used for data handling, pattern matching, environment configuration, and debugging

5. External APIs and Services

The system relies on external APIs to perform advanced processing tasks:

- **Google Translate API:**
Used for translating Sanskrit text into Hindi
- **Google Gemini API:**
Used for AI-based simplification and content generation
- **Tesseract OCR Engine:**
Used for extracting text from images

These services enable the system to perform complex operations without requiring high local computational resources.

6. Development Tools and Environment

- **Anaconda / Virtual Environment:**

Used for managing Python environments and dependencies

- **VS Code / PyCharm:**

Used as Integrated Development Environments (IDEs) for coding and debugging

- **Git (optional):**

Used for version control and project management

7. Web Browser

A modern web browser such as:

- Google Chrome
- Mozilla Firefox
- Microsoft Edge

is required to access and interact with the Streamlit-based frontend interface.

8. Dependency Management

All required libraries and dependencies are maintained using a `requirements.txt` file. This ensures easy installation and reproducibility of the project environment across different systems.

The software requirements for the proposed system are based on widely used and open-source technologies, making the system cost-effective and easy to deploy. The use of Python and modern frameworks ensures flexibility, while integration with external APIs enables advanced AI functionalities without heavy local processing. This combination of tools provides a robust foundation for developing an efficient and scalable learning system.

Chapter 5

METHODOLOGY

5.1 System Architecture

The system architecture of the proposed “**AI-Powered Sanskrit Learning Assistant**” is designed using a modular and layered approach to ensure scalability, maintainability, and efficient processing of data. The architecture integrates multiple components, including frontend, backend, processing modules, and external APIs, to provide a seamless and interactive learning experience.

The system follows a **client-server architecture**, where the frontend handles user interaction and the backend manages data processing and communication with external services. Each component is designed to perform a specific function, contributing to the overall functionality of the system.

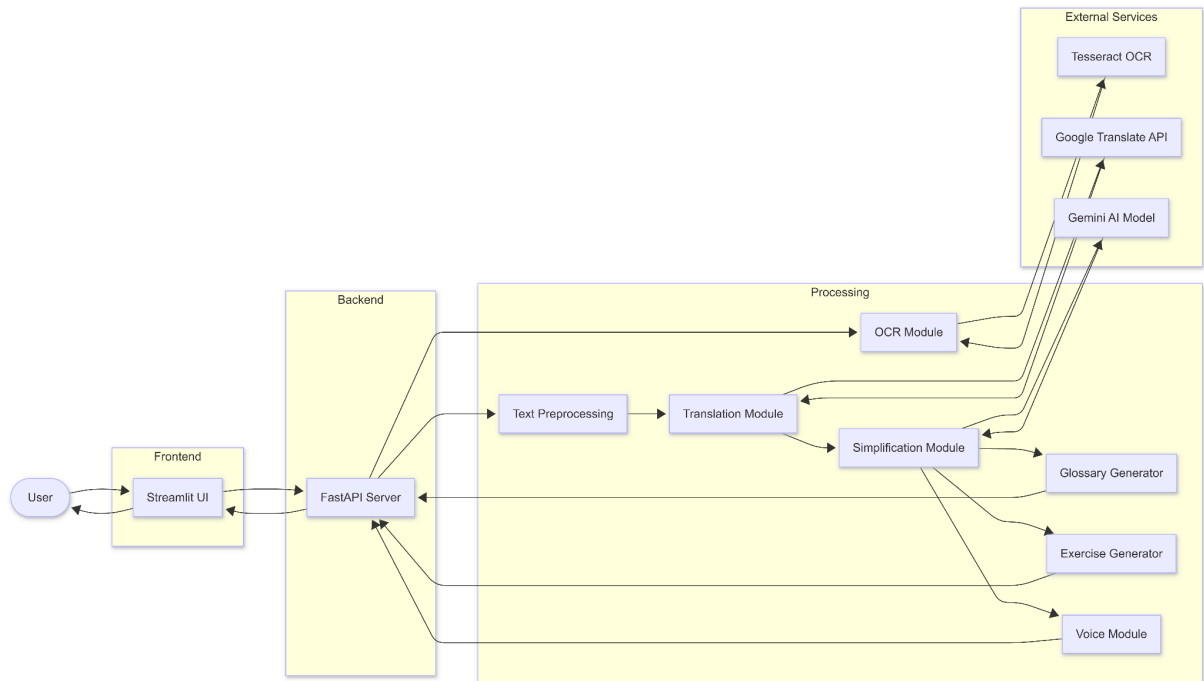


Fig 5.1: System Architecture of AI Sanskrit Learning Assistant

5.1.1 Architectural Overview

The architecture is broadly divided into the following layers:

1. **Presentation Layer (Frontend)**
2. **Application Layer (Backend)**
3. **Processing and Service Layer**
4. **External Services Layer**

These layers interact with each other in a structured manner to process user input and generate meaningful output.

5.1.2 Presentation Layer (Frontend)

The Presentation Layer is responsible for handling all user interactions. It is implemented using **Streamlit**, which provides a simple and interactive web interface.

The key responsibilities of this layer include:

- Accepting user input in the form of text or images
- Displaying extracted Sanskrit text from OCR
- Showing translated and simplified Hindi explanations
- Presenting glossary entries in a structured format
- Rendering interactive modules such as MCQs and fill-in-the-blanks
- Providing audio playback for text-to-speech functionality
- Handling user responses and displaying feedback

The frontend ensures a smooth and intuitive user experience by organizing the workflow into clear steps and visual components.

5.1.3 Application Layer (Backend)

The Application Layer is developed using **FastAPI**, which acts as the core processing unit of the system. It manages API endpoints and coordinates the flow of data between different modules.

The key functions of the backend include:

- Receiving requests from the frontend (OCR, translation, simplification)
- Routing requests to appropriate service modules
- Managing data processing and response generation
- Handling exceptions and errors
- Ensuring efficient communication with external APIs

FastAPI enables asynchronous processing, which improves system performance and reduces response time.

5.1.4 Processing and Service Layer

This layer contains the core functional modules responsible for performing specific operations on the input data. It acts as the bridge between the backend and external services.

The major components of this layer include:

a) OCR Module

- Uses **Tesseract OCR** to extract Sanskrit text from images
- Performs image preprocessing such as resizing and format conversion
- Converts image input into machine-readable text

b) Text Preprocessing Module

- Cleans and normalizes extracted or input text
- Removes unwanted characters and noise
- Ensures proper formatting for further processing

c) Translation Module

- Uses **Google Translate API** to convert Sanskrit text into Hindi
- Acts as an intermediate step before simplification
- Maintains the semantic meaning of the original text

d) Simplification Module

- Utilizes **Gemini API (Large Language Model)**
- Converts complex Hindi into simple, easy-to-understand language
- Generates structured outputs such as simplified text and explanations

e) Glossary Generation Module

- Extracts important Sanskrit terms using rule-based logic
- Generates meanings in simple Hindi
- Structures data into a glossary format

f) Exercise Generation Module

- Creates **MCQs** based on glossary terms
- Generates **fill-in-the-blanks exercises**
- Provides structured question formats for user interaction

g) Voice Processing Module

- **Text-to-Speech (gTTS):** Converts text into audio output
- **Speech Recognition:** Converts user speech into text
- Evaluates spoken responses using matching techniques

h) Evaluation and Feedback Module

- Compares user responses with correct answers
- Provides immediate feedback (correct/incorrect)
- Displays hints and explanations
- Calculates performance scores

5.1.5 External Services Layer

The system integrates several external services to perform advanced operations:

- **Google Translate API:** For language conversion
- **Gemini API:** For AI-based text simplification and content generation
- **Tesseract OCR Engine:** For text extraction from images

These services handle computationally intensive tasks, reducing the load on the local system and improving overall efficiency.

5.1.6 Data Flow Between Components

The data flow in the system follows a sequential pattern:

1. User provides input through the frontend
2. Input is sent to the backend via API requests
3. Backend routes the request to appropriate service modules
4. Processing modules perform OCR, translation, and simplification
5. Results are returned to the frontend
6. Frontend displays output and enables interaction

This structured flow ensures smooth communication and efficient processing at each stage.

5.1.7 Key Characteristics of the Architecture

- **Modularity:** Each component performs a specific function independently
- **Scalability:** New features can be added without affecting existing modules
- **Flexibility:** Supports multiple input formats and learning features
- **Efficiency:** Uses external APIs for heavy processing
- **User-Centric Design:** Focuses on simplicity and ease of use

The system architecture of the proposed solution is designed to efficiently integrate multiple technologies into a unified platform. By combining frontend interaction, backend processing, service modules, and external APIs, the architecture ensures smooth operation and high performance. This structured design supports the overall objective of providing an intelligent and interactive Sanskrit learning system.

5.2 Work flow

The workflow of the proposed system, “**AI-Powered Sanskrit Learning Assistant,**” describes the step-by-step sequence of operations performed from the moment a user provides input to the final generation of simplified output and interactive learning features. The workflow is designed to ensure smooth data processing, accurate results, and an engaging user experience.

The system follows a structured pipeline in which each stage processes the input and passes the result to the next stage, ensuring consistency and efficiency throughout the process.

5.2.1 Overview of Workflow

The workflow consists of the following major stages:

1. Input Acquisition
2. Text Extraction (if image input)
3. Text Preprocessing
4. Translation (Sanskrit → Hindi)
5. AI-Based Simplification
6. Glossary Generation
7. Interactive Learning Module Generation
8. Voice Processing (Optional)
9. Evaluation and Feedback
10. Output Display and Storage

Each stage plays a critical role in transforming raw input into meaningful and interactive output.

5.2.2 Detailed Step-by-Step Workflow

Step 1: Input Acquisition

The workflow begins with the user providing Sanskrit input through the frontend interface. The system supports two types of input:

- Uploading an image containing Sanskrit text
- Entering Sanskrit text manually in Devanagari script

This flexibility allows users to interact with the system based on their convenience and available resources.

Step 2: Text Extraction (OCR Processing)

If the input is provided as an image, the system applies Optical Character Recognition (OCR) using Tesseract to extract the textual content.

- The image is preprocessed (resizing, format conversion)
- Noise and distortions are minimized
- Sanskrit text is extracted and converted into digital format

If the input is already in text form, this step is skipped.

Step 3: Text Preprocessing and Cleaning

The extracted or manually entered text is processed to ensure quality and consistency. This includes:

- Unicode normalization
- Removal of unwanted characters or noise
- Formatting corrections

This step ensures that the text is clean and ready for further processing.

Step 4: Translation (Sanskrit to Hindi)

The cleaned Sanskrit text is passed to the translation module, which uses the Google Translate API to convert it into Hindi.

- The translation maintains the original meaning
- It acts as a base for simplification
- Provides an intermediate understandable representation

Step 5: AI-Based Simplification

The translated Hindi text is then processed using a Large Language Model (Gemini API) to generate simplified explanations.

- Complex sentences are broken into simpler forms
- Difficult vocabulary is replaced with easy words
- Meaning is preserved while improving clarity

This step ensures that the output is suitable for school-level students.

Step 6: Glossary Generation

The system extracts important Sanskrit terms from the original input using rule-based techniques.

- Key terms are identified
- Meanings are generated in simple Hindi
- Results are structured into a glossary format

This supports vocabulary development and deeper understanding.

Step 7: Interactive Learning Module Generation

Based on the glossary and simplified content, the system generates interactive learning features:

- **Multiple Choice Questions (MCQs):**
Questions are generated to test understanding of meanings
- **Fill-in-the-Blanks Exercises:**
Sentences are created with missing words for practice

These features encourage active learning and self-assessment.

Step 8: Voice Processing (Optional)

The system includes optional voice-based features to enhance accessibility:

- **Text-to-Speech (TTS):**
Converts simplified Hindi text into audio for listening
- **Speech Recognition:**
Captures user speech and converts it into text

This enables auditory learning and pronunciation practice.

Step 9: Evaluation and Feedback

The system evaluates user responses in exercises and provides immediate feedback:

- Identifies correct and incorrect answers
- Displays correct answers with explanations
- Provides hints where necessary
- Calculates scores for performance tracking

This helps users learn from mistakes and improve understanding.

Step 10: Output Display and Storage

Finally, the system displays the results to the user in a structured format:

- Simplified Hindi explanation
- Glossary of important terms
- Interactive exercises and results

Additionally, the system stores the data in history for future reference and revision.

5.2.3 Workflow Characteristics

The workflow of the system exhibits the following characteristics:

- **Sequential Processing:** Each stage builds upon the previous stage
- **Flexibility:** Supports multiple input types and optional features
- **Efficiency:** Uses APIs for fast and accurate processing
- **Interactivity:** Includes user engagement at multiple stages
- **Scalability:** Can be extended with additional features

5.2.4 Summary

The workflow of the proposed system ensures a systematic transformation of Sanskrit input into simplified and interactive output. By integrating multiple stages such as OCR, translation, AI-based simplification, and user interaction, the system provides a comprehensive learning solution. This structured workflow enhances usability, improves learning efficiency, and supports better understanding of Sanskrit texts.

Chapter 6

FLOW DIAGRAMS

6.1 Data flow diagrams

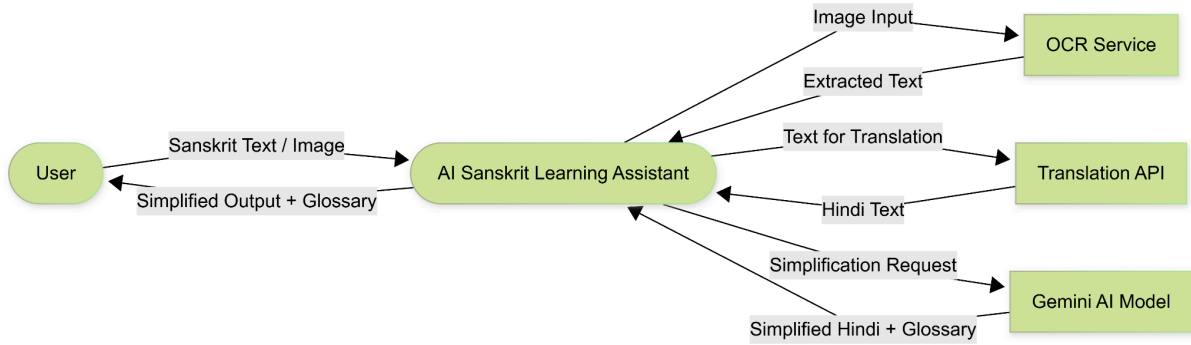


Fig 6.1.1: Level 0 Data Flow Diagram of AI Sanskrit Learning Assistant

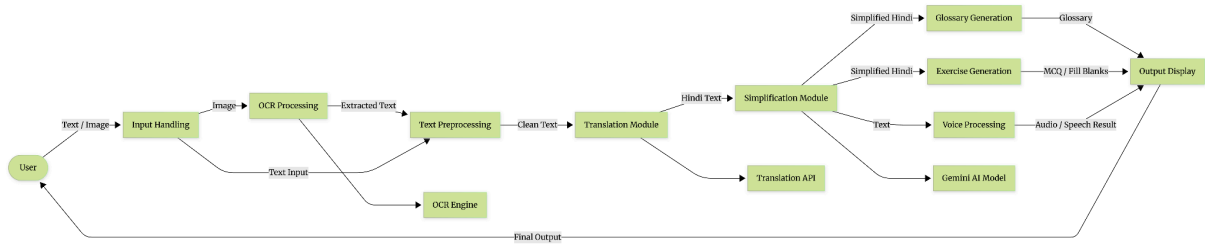


Fig 6.1.2: Level 1 Data Flow Diagram of AI Sanskrit Learning Assistant

6.2 Sequential UML Diagrams

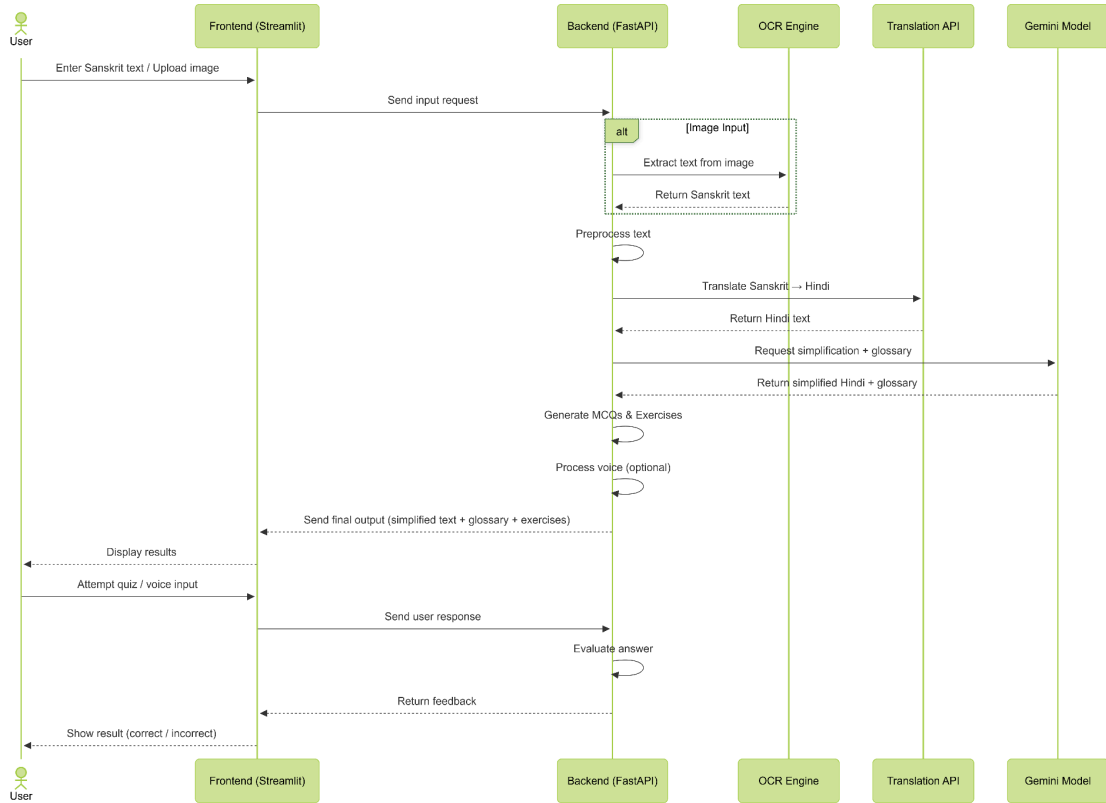


Fig 6.2: Sequence Diagram of AI Sanskrit Learning Assistant

Chapter 7

IMPLEMENTATION

7.1 Code Snippet

1. API Endpoint (Core Backend Entry Point)

```
@app.post("/api/simplify")
def simplify_text(data: RequestModel):
    result = simplify_with_glossary(data.sanskrit, data.hindi)
    return result
```

Explanation:

This endpoint receives Sanskrit and Hindi input from the frontend and processes it using the core simplification service.

2. Sanskrit Term Extraction Logic

```
tokens = re.findall(r"[\u0900-\u097F]{2,}", text)

terms = []
for t in tokens:
    if t not in ["I", "II"] and len(t) > 2:
        terms.append(t)
```

Explanation:

This logic extracts meaningful Sanskrit words using Unicode filtering and removes unwanted symbols.

3. AI Simplification (Gemini API Call)

```
response = model.generate_content(
    prompt,
    generation_config={"temperature": 0.2}
)
```

```
simplified_text = response.text.strip()
```

Explanation:

This module sends the Hindi text to the Gemini model and generates a simplified version suitable for students.

4. Glossary Generation Logic

```
terms_text = ", ".join(terms)
```

```
prompt = f"Explain these Sanskrit terms in simple Hindi: {terms_text}"
```

```
response = model.generate_content(prompt)
```

Explanation:

Important Sanskrit terms are passed to the AI model to generate simple meanings for glossary creation.

5. Frontend API Call (Connecting UI to Backend)

```
payload = {"sanskrit": sanskrit, "hindi": hindi}
```

```
response = requests.post(SIMPLIFY_URL, json=payload)
```

```
return response.json()
```

Explanation:

This connects the frontend interface with the backend API to fetch processed results.

6. MCQ Generation Logic

```
correct = word["meaning"]
```

```
options = random.sample(all_meanings, 3)
```

```
options.append(correct)
```

```
random.shuffle(options)
```

Explanation:

This logic generates multiple-choice questions by mixing correct and incorrect options.

7. Voice Feature (Text-to-Speech)

```
from gtts import gTTS
```

```
tts = gTTS(text=simplified_text, lang="hi")
```

```
tts.save("output.mp3")
```

Explanation:

This converts simplified Hindi text into audio for better accessibility and learning.

7.2 Screen shots& Results

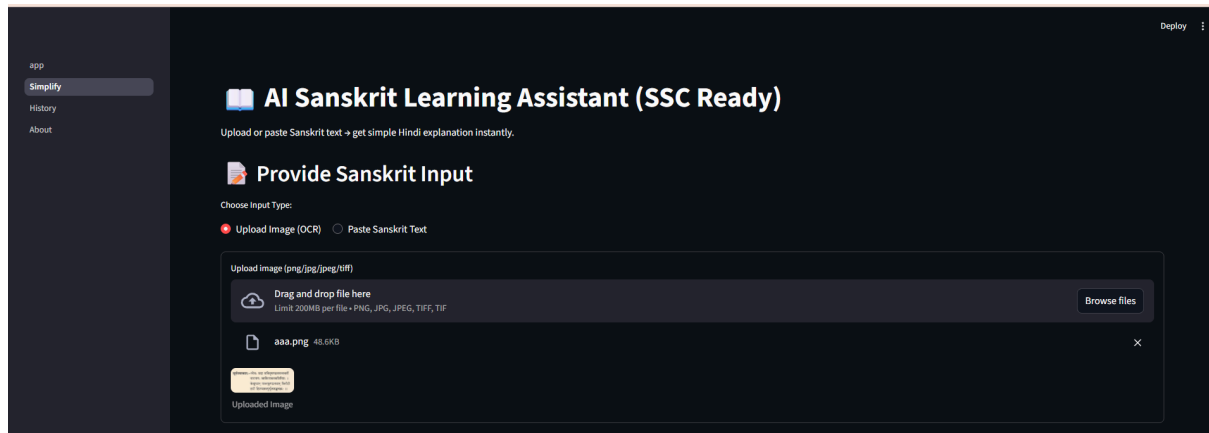


Fig 7.2.1: User Input Interface for Sanskrit Text

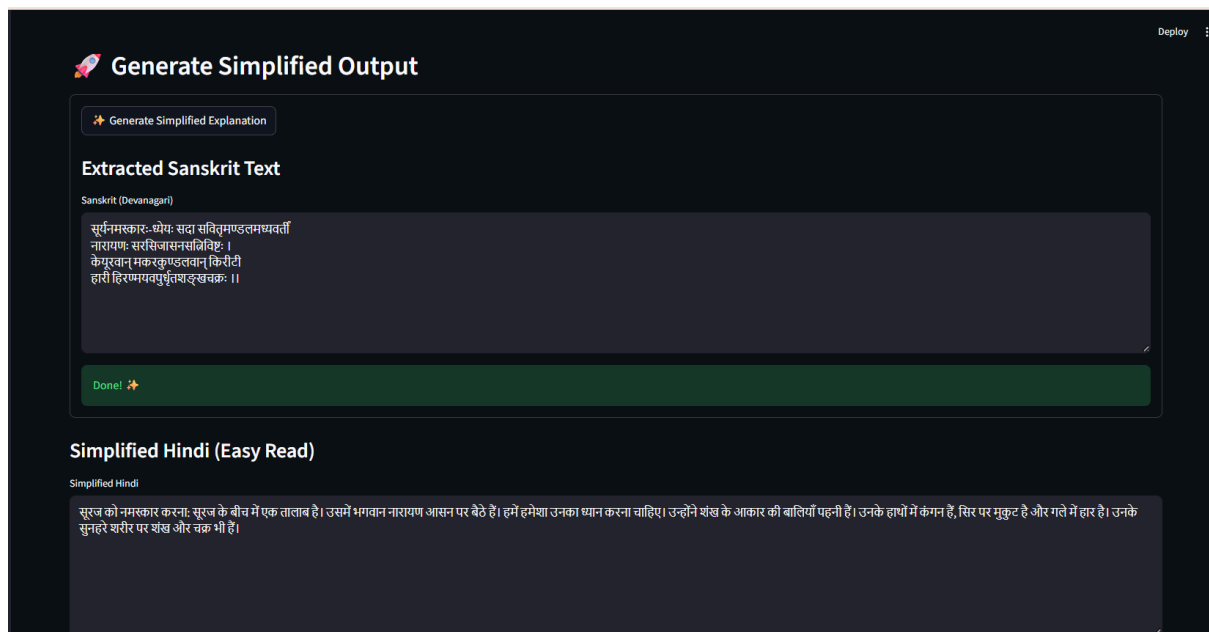


Fig 7.2.2: Extracted Sanskrit Text using OCR and Simplified Hindi Output

Glossary (Sanskrit → Simple Hindi)		
	word	meaning
0	सूर्यमस्कारः	यह सूर्य को प्रणाम करने की एक विधि है।
1	ध्येयः	जिस पर ध्यान केंद्रित किया जाता है।
2	सदा	हमेशा या निरंतर।
3	सवित्रमण्डलमध्यवर्ती	सूर्य के मंडल के बीच में स्थित।
4	नासयणः	यह भगवान विष्णु का एक नाम है।
5	सरसिंहासनसज्जिविष्टः	कमल के आसन पर विराजमान।
6	केयूरवान्	जिनके हाथों में बाज्रबंद हों।
7	मकरकुण्डलवान्	जिनके कानों में मगरमच्छ के आकार के कुंडल हों।
8	किरीटी	जिनके सिर पर मुकुट हों।
9	हारी	जो मन को हर ले या बहुत सुंदर हों।

Fig 7.2.3: Generated Glossary Output

Listen Explanation

Play Audio

Speak & Verify Exercise

Speak the meaning of: ध्येयः

Start Recording

You said: जिस पर ध्यान केंद्रित किया जाता है

Correct! Well done 🎉

Try Another Word

Fig 7.2.4: Voice-Based Learning and Evaluation

Practice MCQs

Generate MCQs

Q1. 'हमेशा या निरंतर' के लिए सही संस्कृत शब्द चुनिए।

Choose answer:

☐ सवित्रमण्डलमध्यवर्ती

☒ सदा

☐ हिरण्यमयवपुर्धृतशङ्खचक्रः

☐ हारी

Q2. 'यह भगवान विष्णु का एक नाम है।' के लिए सही संस्कृत शब्द चुनिए।

Choose answer:

☒ हारी

☐ सरसिंहासनसज्जिविष्टः

☐ नासयणः

☐ केयूरवान्

Q3. 'जिनके सिर पर मुकुट हों।' के लिए सही संस्कृत शब्द चुनिए।

Choose answer:

☒ सवित्रमण्डलमध्यवर्ती

☐ हारी

☐ सरसिंहासनसज्जिविष्टः

☐ किरीटी

Q4. 'सूर्यमस्कारः' का अर्थ क्या है?

Choose answer:

☒ जिस पर ध्यान केंद्रित किया जाता है।

Fig 7.2.5 (a): Multiple Choice Question (MCQ) Module

☐ जो मन को हर ले या बहुत सुंदर हो।
☒ यह सूर्य को प्रणाम करने की एक विधि है।
☐ जिनके हाथों में बाजूबंद हों।

Q5. 'जिनके हाथों में बाजूबंद हों।' के लिए सही संस्कृत शब्द चुनिए।

Choose answer:

☐ हिरण्यपवपुर्भूतशङ्खचक्रः
☐ मकरकुण्डलवान्
☐ सूर्यमस्कारः
☒ केयूरवान्

☒ Submit Answers

Q1: Correct ✓

Q2: Wrong ✗ (Correct: नारायणः)

Q3: Wrong ✗ (Correct: किरीटी)

Q4: Correct ✓

Q5: Correct ✓

Fig 7.2.5 (b): Multiple Choice Question (MCQ) Module

 **Fill in the Blanks**

☒ Generate Fill in the Blanks

Shloka

_____ (यह सूर्य को प्रणाम करने की एक विधि है।)। ध्येयः _____ (हमेशा या नित्य।) सवितुर्मण्डलमध्यवर्ती
 नारायणः _____ (कमल के आसन पर विराजमान।)।
 केयूरवान् मकरकुण्डलवान् किरीटी
 _____ (जो मन को हर ले या बहुत सुंदर हो।)। हिरण्यपवपुर्भूतशङ्खचक्रः ॥

Blank 1: सूर्यमस्कारः

Blank 2: सदा

Blank 3: सरसिजासनसवितुः

Blank 4: हारी

☒ Check Answers

Blank 1: Correct ✓

Blank 2: Correct ✓

Blank 3: Correct ✓

Fig 7.2.6: Fill in the Blanks Exercise

Chapter 8

Testing Report

8.1 Functional Test Cases

Test Case ID	Module	Input Type	Test Description	Expected Output	Actual Output	Status
TC01	Input Module	Sanskrit Text	Enter valid Sanskrit text	Text accepted and processed	Text processed successfully	Pass
TC02	OCR Module	Image Input	Upload Sanskrit image	Correct text extracted	Text extracted correctly	Pass
TC03	Preprocessing	Raw Text	Clean and normalize text	Cleaned Sanskrit text	Text normalized	Pass
TC04	Translation Module	Sanskrit Text	Translate Sanskrit to Hindi	Correct Hindi translation	Accurate Hindi output	Pass
TC05	Simplification Module	Hindi Text	Simplify Hindi text	Easy, readable Hindi	Simplified correctly	Pass

TC06	Glossary Generation	Sanskrit Text	Extract important terms	Relevant Sanskrit terms identified	Terms extracted properly	Pass
TC07	Glossary Output	Terms List	Generate meanings	Correct meanings in simple Hindi	Meanings generated correctly	Pass
TC08	MCQ Generation	Glossary Data	Generate MCQs	Valid question with 4 options	MCQs generated correctly	Pass
TC09	MCQ Evaluation	User Selection	Validate selected option	Correct/Incorrect feedback	Feedback displayed correctly	Pass
TC10	Fill in Blanks	Sentence + Options	Fill blanks using options	Correct placement of words	Words placed correctly	Pass
TC11	Fill Evaluation	User Answers	Validate blank answers	Correct score and feedback	Evaluation accurate	Pass
TC12	Voice (TTS)	Hindi Text	Convert text to speech	Audio generated	Audio played successfully	Pass
TC13	Voice (STT)	Speech Input	Convert speech to text	Recognized text	Speech recognized correctly	Pass

TC14	Voice Evaluation	Spoken Answer	Match spoken response	Correct/Incorrect feedback	Matching working correctly	Pass
TC15	API Integration	API Request	Backend communication	JSON response	Response received correctly	Pass

8.2 Performance Test Cases

Test Case ID	Scenario	Input Size	Expected Result	Actual Result	Status
TC16	Small Input	1–2 lines	Fast response (< 3 sec)	Response within limit	Pass
TC17	Medium Input	Paragraph	Moderate response time	Acceptable delay	Pass
TC18	Large Input	Long text	Slight delay but correct output	Output generated correctly	Pass
TC19	Multiple Requests	Repeated inputs	System stability	No crash observed	Pass

8.3 Error Handling Test Cases

Test Case ID	Scenario	Input	Expected Output	Actual Output	Status
TC20	Empty Input	Blank	Error message	Proper warning shown	Pass
TC21	Invalid Image	Corrupt file	Error handling	Error handled	Pass

TC22	API Failure	No internet	Graceful failure message	Error message displayed	Pass
TC23	Incorrect Voice Input	Noise speech	Low accuracy but handled	System handled input	Pass

8.4 Summary Table

Category	Total Cases	Passed	Failed
Functional Testing	15	15	0
Performance Testing	4	4	0
Error Handling	4	4	0
Total	23	23	0

9. CONCLUSION

The project “**AI-Powered Sanskrit Learning Assistant**” successfully addresses the challenges associated with understanding Sanskrit texts by leveraging modern Artificial Intelligence and Natural Language Processing techniques. The system provides an effective solution for simplifying complex Sanskrit content into easy and understandable Hindi, thereby bridging the gap between classical language and modern learners.

By integrating technologies such as Optical Character Recognition (OCR), machine translation, and Large Language Models, the system automates the process of text extraction, translation, and simplification. In addition to this, the inclusion of interactive learning features such as glossary generation, Multiple Choice Questions, fill-in-the-blanks, and voice-based interaction enhances user engagement and promotes active learning.

The system has been designed with a user-friendly interface and modular architecture, ensuring ease of use, scalability, and efficient performance. The testing results demonstrate that the system functions reliably and meets its intended objectives.

Overall, the project shifts the learning approach from memorization to conceptual understanding, making Sanskrit more accessible and engaging for students. It demonstrates the potential of AI-driven solutions in modernizing traditional education and improving learning outcomes.

10. FUTURE WORK

The proposed system, “**AI-Powered Sanskrit Learning Assistant,**” provides a strong foundation for simplifying Sanskrit learning using Artificial Intelligence. However, there are several opportunities to enhance the system further by incorporating additional features, improving accuracy, and expanding its capabilities. The following points outline the potential future improvements of the system:

1. Multilingual Support

Currently, the system primarily focuses on simplifying Sanskrit text into Hindi. In the future, it can be extended to support multiple regional and international languages such as English, Marathi, and others. This would make the system more accessible to a wider audience and enhance its usability across different linguistic backgrounds.

2. Improved OCR Accuracy

Although the system uses OCR for extracting text from images, the accuracy may vary depending on image quality. Future enhancements can include:

- Advanced image preprocessing techniques
- Integration with more sophisticated OCR models
- Support for handwritten Sanskrit text

This will improve text extraction accuracy and reliability.

3. Enhanced AI-Based Simplification

The simplification process can be further improved by:

- Training domain-specific language models for Sanskrit literature
- Providing context-aware explanations based on chapter or topic
- Allowing customization of explanation levels (easy, medium, advanced)

This will make the system more adaptive to different learning needs.

4. Advanced Voice Interaction

The current voice module can be expanded to include:

- Real-time pronunciation feedback
- Voice-based navigation of the system
- Conversational chatbot-style interaction

This would create a more immersive and interactive learning experience.

5. Personalized Learning System

Future versions of the system can incorporate personalization features such as:

- Tracking user progress and performance
- Recommending exercises based on weak areas
- Adaptive difficulty levels for quizzes

This will enable a more customized learning experience for each user.

6. Mobile Application Development

The system can be extended into a mobile application (Android/iOS) to increase accessibility and usability. A mobile platform would allow students to learn anytime and anywhere, making the system more practical for real-world usage.

7. Integration with Educational Platforms

The system can be integrated with existing e-learning platforms or school management systems. This would allow teachers to:

- Assign exercises
- Monitor student progress
- Use the tool as a teaching aid in classrooms

8. Gamification of Learning

To further improve engagement, gamification features can be added, such as:

- Reward points and badges
- Leaderboards
- Level-based progression

This will make learning more enjoyable and motivating for students.

9. Improved Evaluation and Analytics

Future improvements can include detailed performance analytics such as:

- Accuracy tracking over time
- Topic-wise performance analysis
- Visual dashboards for progress monitoring

This will help students and teachers better understand learning patterns.

10. Offline Functionality

Currently, the system relies on internet connectivity for API-based processing. Future versions can explore:

- Offline capabilities using lightweight models
- Local caching of frequently used data

This will improve accessibility in low-connectivity environments.

11. Expansion to Other Subjects

The framework of the system can be extended beyond Sanskrit to other subjects such as:

- Hindi literature
- English comprehension
- Historical texts

This will increase the scope and applicability of the system.

The proposed system has significant potential for further enhancement and expansion. By incorporating advanced technologies, improving interactivity, and expanding its scope, the system can evolve into a comprehensive educational platform. These future improvements will make the system more robust, scalable, and effective in supporting modern learning needs.

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